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True size eyewear in real time

Abstract

Methods and systems are disclosed for performing operations comprising: receiving an image that includes a depiction of a face of a user; generating a plurality of landmarks of the face based on the received image; removing a set of interfering landmarks from the plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks; obtaining a depth map for the face of the user; and computing a real-world scale of the face of the user based on the depth map and the remaining set of landmarks.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a continuation of U.S. patent application Ser. No. 17/208,159, filed on Mar. 22, 2021, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to providing augmented reality experiences using a messaging application.

BACKGROUND

(2) Augmented-Reality (AR) is a modification of a virtual environment. For example, in Virtual Reality (VR), a user is completely immersed in a virtual world, whereas in AR, the user is immersed in a world where virtual objects are combined or superimposed on the real world. An AR system aims to generate and present virtual objects that interact realistically with a real-world environment and with each other. Examples of AR applications can include single or multiple player video games, instant messaging systems, and the like.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

(2) FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

(3) FIG. 2 is a diagrammatic representation of a messaging client application, in accordance with some examples.

(4) FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

(5) FIG. 4 is a diagrammatic representation of a message, in accordance with some examples.

(6) FIG. 5 is a block diagram showing an example true size estimation system, according to example examples.

(7) FIGS. 6-9 are diagrammatic representations of outputs of the true size estimation system, in

accordance with some examples.

(8) FIGS. **10A** and **10B** are flowcharts illustrating example operations of the messaging application server, according to examples.

(9) FIG. **11** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

(10) FIG. **12** is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

(11) The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples. It will be evident, however, to those skilled in the art, that examples may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

(12) Typically, virtual reality (VR) and augmented reality (AR) systems allow users to add augmented reality elements, such as augmented reality glasses, to a face of the user depicted in a captured image. To do so, the typical VR/AR systems use specialized techniques that require calibration to determine a scale of the user's face in the image. For example, such systems instruct the user to place a reference object, such as a credit card, on the user's face or next to the user's face so that a scale of the face can be computed. The systems can then display augmented reality glasses on the user's face based on the calibration. While such systems generally work well, the need to calibrate the systems places an additional burden on the users and takes away from the enjoyment of the experience. Also, computing the scale by calibrating the system takes additional time and resources, making such systems less efficient for general applications.

(13) The disclosed techniques improve the efficiency of using an electronic device which implements or otherwise accesses an AR/VR system by computing a true or real-world scale of a user's face by combining a select set of facial landmarks with a depth map of the user's face. Specifically, the disclosed techniques receive an image that includes a depiction of a face of a user and generate a plurality of facial landmarks based on the received image. Facial landmarks (or landmarks on the face) can correspond to a predefined region of a person's face such as a nose, mouth, eyes, etc. The disclosed techniques remove a set of interfering facial landmarks from the plurality of facial landmarks resulting in a remaining set of landmarks of the plurality of landmarks. The disclosed techniques obtain a depth map for the face of the user and compute a real-world scale of the face of the user based on the depth map and the remaining set of landmarks. The real-world scale of the face is then used to adjust a size of an augmented reality element, such as augmented reality glasses (e.g., eyewear) or an augmented reality hat. The real-world scale together with a facial topology is also used to identify the appropriate position over which to add or display the augmented reality element on the user's face. As the user moves the face around in a video, the positioning of the augmented reality element continues to be changed to remain fixed to the identified position of the face. The real-world scale of the face continues to be updated as new images of a video depicting the user's face are received and processed in a similar manner.

(14) In this way, the disclosed techniques can apply one or more visual effects to the user's face in the current image without performing any calibration operations or pre-capture operations. This improves the overall experience of the user in using the electronic device and reduces the overall amount of system resources needed to accomplish a task.

(15) Networked Computing Environment

(16) FIG. **1** is a block diagram showing an example messaging system **100** for exchanging data (e.g., messages and associated content) over a network. The messaging system **100** includes multiple instances of a client device **102**, each of which hosts a number of applications, including a

messaging client **104** and other external applications **109** (e.g., third-party applications). Each messaging client **104** is communicatively coupled to other instances of the messaging client **104** (e.g., hosted on respective other client devices **102**), a messaging server system **108** and external app(s) servers **110** via a network **112** (e.g., the Internet). A messaging client **104** can also communicate with locally-hosted third-party applications **109** using Applications Program Interfaces (APIs).

(17) A messaging client **104** is able to communicate and exchange data with other messaging clients **104** and with the messaging server system **108** via the network **112**. The data exchanged between messaging clients **104**, and between a messaging client **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

(18) The messaging server system **108** provides server-side functionality via the network **112** to a particular messaging client **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client **104** or by the messaging server system **108**, the location of certain functionality either within the messaging client **104** or the messaging server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108** but to later migrate this technology and functionality to the messaging client **104** where a client device **102** has sufficient processing capacity.

(19) The messaging server system **108** supports various services and operations that are provided to the messaging client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client **104**.

(20) Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **116** is coupled to, and provides a programmatic interface to, application servers **114**. The application servers **114** are communicatively coupled to a database server **120**, which facilitates access to a database **126** that stores data associated with messages processed by the application servers **114**. Similarly, a web server **128** is coupled to the application servers **114**, and provides web-based interfaces to the application servers **114**. To this end, the web server **128** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

(21) The Application Program Interface (API) server **116** receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application servers **114**. Specifically, the Application Program Interface (API) server **116** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client **104** in order to invoke functionality of the application servers **114**. The Application Program Interface (API) server **116** exposes various functions supported by the application servers **114**, including account registration, login functionality, the sending of messages, via the application servers **114**, from a particular messaging client **104** to another messaging client **104**, the sending of media files (e.g., images or video) from a messaging client **104** to a messaging server **118**, and for possible access by another messaging client **104**, the settings of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph), the location of friends within a social graph, and opening an application event (e.g., relating to the messaging client **104**).

(22) The application servers **114** host a number of server applications and subsystems, including for example a messaging server **118**, an image processing server **122**, and a social network server **124**.

The messaging server **118** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client **104**. Other processor- and memory-intensive processing of data may also be performed server-side by the messaging server **118**, in view of the hardware requirements for such processing.

(23) The application servers **114** also include an image processing server **122** that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server **118**.

(24) Image processing server **122** is used to implement scan functionality of the augmentation system **208**. Scan functionality includes activating and providing one or more augmented reality experiences on a client device **102** when an image is captured by the client device **102**. Specifically, the messaging client **104** on the client device **102** can be used to activate a camera. The camera displays one or more real-time images or a video to a user along with one or more icons or identifiers of one or more augmented reality experiences. The user can select a given one of the identifiers to launch the corresponding augmented reality experience or perform a desired image modification (e.g., adding augmented reality eyewear or a hat to a face depicted in an image).

(25) The social network server **124** supports various social networking functions and services and makes these functions and services available to the messaging server **118**. To this end, the social network server **124** maintains and accesses an entity graph **308** (as shown in FIG. 3) within the database **126**. Examples of functions and services supported by the social network server **124** include the identification of other users of the messaging system **100** with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.

(26) Returning to the messaging client **104**, features and functions of an external resource (e.g., a third-party application **109** or applet) are made available to a user via an interface of the messaging client **104**. The messaging client **104** receives a user selection of an option to launch or access features of an external resource (e.g., a third-party resource), such as external apps **109**. The external resource may be a third-party application (external apps **109**) installed on the client device **102** (e.g., a “native app”), or a small-scale version of the third-party application (e.g., an “applet”) that is hosted on the client device **102** or remote of the client device **102** (e.g., on third-party servers **110**). The small-scale version of the third-party application includes a subset of features and functions of the third-party application (e.g., the full-scale, native version of the third-party standalone application) and is implemented using a markup-language document. In one example, the small-scale version of the third-party application (e.g., an “applet”) is a web-based, markup-language version of the third-party application and is embedded in the messaging client **104**. In addition to using markup-language documents (e.g., a *.ml file), an applet may incorporate a scripting language (e.g., a *.js file or a .json file) and a style sheet (e.g., a *.ss file).

(27) In response to receiving a user selection of the option to launch or access features of the external resource (external app **109**), the messaging client **104** determines whether the selected external resource is a web-based external resource or a locally-installed external application. In some cases, external applications **109** that are locally installed on the client device **102** can be launched independently of and separately from the messaging client **104**, such as by selecting an icon, corresponding to the external application **109**, on a home screen of the client device **102**. Small-scale versions of such external applications can be launched or accessed via the messaging client **104** and, in some examples, no or limited portions of the small-scale external application can be accessed outside of the messaging client **104**. The small-scale external application can be launched by the messaging client **104** receiving, from an external app(s) server **110**, a markup-

language document associated with the small-scale external application and processing such a document.

(28) In response to determining that the external resource is a locally-installed external application **109**, the messaging client **104** instructs the client device **102** to launch the external application **109** by executing locally-stored code corresponding to the external application **109**. In response to determining that the external resource is a web-based resource, the messaging client **104** communicates with the external app(s) servers **110** to obtain a markup-language document corresponding to the selected resource. The messaging client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client **104**.

(29) The messaging client **104** can notify a user of the client device **102**, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the messaging client **104** can provide participants in a conversation (e.g., a chat session) in the messaging client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using a respective messaging client **104**, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

(30) The messaging client **104** can present a list of the available external resources (e.g., third-party or external applications **109** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the external application **109** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

(31) System Architecture

(32) FIG. 2 is a block diagram illustrating further details regarding the messaging system **100**, according to some examples. Specifically, the messaging system **100** is shown to comprise the messaging client **104** and the application servers **114**. The messaging system **100** embodies a number of subsystems, which are supported on the client side by the messaging client **104** and on the sever side by the application servers **114**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **208**, a map system **210**, a game system **212**, and an external resource system **220**.

(33) The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the messaging client **104** and the messaging server **118**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

(34) The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection

management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client **104**.

(35) The collection management system **204** furthermore includes a curation interface **206** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **206** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.

(36) The augmentation system **208** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system **208** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The augmentation system **208** operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client **104** based on a geolocation of the client device **102**. In another example, the augmentation system **208** operatively supplies a media overlay to the messaging client **104** based on other information, such as social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay may include text, a graphical element, or image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **208** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **126** and accessed through the database server **120**.

(37) In some examples, the augmentation system **208** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **208** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

(38) In other examples, the augmentation system **208** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **208** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time. The augmentation system **208** communicates with the image processing server **122** to obtain augmented reality experiences and presents identifiers of such experiences in one or more user interfaces (e.g., as icons over a real-time image or video or as thumbnails or icons in interfaces dedicated for presented identifiers of augmented reality experiences). Once an augmented reality experience is selected, one or more images, videos, or augmented reality graphical elements are retrieved and presented as an overlay on top of the images or video captured by the client device **102**. In some cases, the camera is switched to a front-facing view (e.g., the front-facing camera of the client device **102** is activated in response to activation of a particular augmented reality experience) and the images from the front-facing camera of the client device **102** start being displayed on the client device **102** instead of the rear-facing camera of the client device **102**. The one or more images, videos, or augmented reality graphical elements are retrieved and presented as

an overlay on top of the images that are captured and displayed by the front-facing camera of the client device **102**.

(39) In other examples, the augmentation system **208** is able to communicate and exchange data with another augmentation system **208** on another client device **102** and with the server via the network **106**. The data exchanged can include a session identifier that identifies the shared AR session, a transformation between a first client device **102** and a second client device **102** (e.g., a plurality of client devices **102** include the first and second devices) that is used to align the shared AR session to a common point of origin, a common coordinate frame, functions (e.g., commands to invoke functions) as well as other payload data (e.g., text, audio, video or other multimedia data).

(40) The augmentation system **208** sends the transformation to the second client device **102** so that the second client device **102** can adjust the AR coordinate system based on the transformation. In this way, the first and second client devices **102** synch up their coordinate systems and frames for displaying content in the AR session. Specifically, the augmentation system **208** computes the point of origin of the second client device **102** in the coordinate system of the first client device **102**. The augmentation system **208** can then determine an offset in the coordinate system of the second client device **102** based on the position of the point of origin from the perspective of the second client device **102** in the coordinate system of the second client device **102**. This offset is used to generate the transformation so that the second client device **102** generates AR content in according to a common coordinate system or frame as the first client device **102**.

(41) The augmentation system **208** that can communicate with the client device **102** to establish individual or shared AR sessions. The augmentation system **208** can also be coupled to the messaging server **118** to establish an electronic group communication session (e.g., group chat, instant messaging) for the client devices **102** in a shared AR session. The electronic group communication session can be associated with a session identifier provided by the client devices **102** to gain access to the electronic group communication session and to the shared AR session. In one example, the client devices **102** first gain access to the electronic group communication session and then obtain the session identifier in the electronic group communication session that allows the client devices **102** to access to the shared AR session. In some examples, the client devices **102** are able to access the shared AR session without aid or communication with the augmentation system **208** in the application servers **114**.

(42) The map system **210** provides various geographic location functions, and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **210** enables the display of user icons or avatars (e.g., stored in profile data **316**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

(43) The game system **212** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games (e.g., web-based games or web-based applications) that can be launched by a user within the context of the messaging client **104**, and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-

game rewards (e.g., coins and items).

(44) The external resource system **220** provides an interface for the messaging client **104** to communicate with external app(s) servers **110** to launch or access external resources. Each external resource (apps) server **110** hosts, for example, a markup language (e.g., HTML5) based application or small-scale version of an external application (e.g., game, utility, payment, or ride-sharing application that is external to the messaging client **104**). The messaging client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the external resource (apps) servers **110** associated with the web-based resource. In certain examples, applications hosted by external resource servers **110** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the messaging server **118**. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. In certain examples, the messaging server **118** includes a JavaScript library that provides a given third-party resource access to certain user data of the messaging client **104**. HTML5 is used as an example technology for programming games, but applications and resources programmed based on other technologies can be used.

(45) In order to integrate the functions of the SDK into the web-based resource, the SDK is downloaded by an external resource (apps) server **110** from the messaging server **118** or is otherwise received by the external resource (apps) server **110**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the messaging client **104** into the web-based resource.

(46) The SDK stored on the messaging server **118** effectively provides the bridge between an external resource (e.g., third-party or external applications **109** or applets and the messaging client **104**). This provides the user with a seamless experience of communicating with other users on the messaging client **104**, while also preserving the look and feel of the messaging client **104**. To bridge communications between an external resource and a messaging client **104**, in certain examples, the SDK facilitates communication between external resource servers **110** and the messaging client **104**. In certain examples, a Web ViewJavaScriptBridge running on a client device **102** establishes two one-way communication channels between a external resource and the messaging client **104**. Messages are sent between the external resource and the messaging client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

(47) By using the SDK, not all information from the messaging client **104** is shared with external resource servers **110**. The SDK limits which information is shared based on the needs of the external resource. In certain examples, each external resource server **110** provides an HTML5 file corresponding to the web-based external resource to the messaging server **118**. The messaging server **118** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the messaging client **104**. Once the user selects the visual representation or instructs the messaging client **104** through a GUI of the messaging client **104** to access features of the web-based external resource, the messaging client **104** obtains the HTML5 file and instantiates the resources necessary to access the features of the web-based external resource.

(48) The messaging client **104** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the messaging client **104** determines whether the launched external resource has been previously authorized to access user data of the messaging client **104**. In response to determining that the launched external resource has been previously authorized to access user data of the messaging client **104**, the messaging client **104** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the

messaging client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the messaging client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle of or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the messaging client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the messaging client **104**. In some examples, the external resource is authorized by the messaging client **104** to access the user data in accordance with an OAuth 2 framework.

(49) The messaging client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale external applications (e.g., a third-party or external application **109**) are provided with access to a first type of user data (e.g., only two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of external applications (e.g., web-based versions of third-party applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

(50) The true size estimation system **224** computes a real-world scale of a user's face that is depicted in an image, such as based on facial landmarks (or subset of facial landmarks) and data received from a depth sensor. For example, the true size estimation system **224** can perform object recognition on the captured video feed to generate a plurality of landmarks of the face depicted in a received image. In response to generating the plurality of landmarks, the true size estimation system **224** removes a set of interfering landmarks from the plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks, such as by removing a hair landmark or chin landmark. In some cases, the true size estimation system **224** removes landmarks that have a visibility parameter or stability parameter that is lower than a specified value. In some cases, the true size estimation system **224** sorts the landmarks based on respective visibility and stability parameters to generate the remaining set of landmarks. The true size estimation system **224** obtains a depth map and then computes a real-world scale of the face of the user based on the depth map and the remaining set of landmarks. An illustrative implementation of the true size estimation system **224** is shown and described in connection with FIG. 5 below.

(51) Specifically, the true size estimation system **224** is a component that can be accessed by an AR/VR application implemented on the client device **102**. The AR/VR application uses an RGB camera to capture a monocular image of a user's real-world face. The AR/VR application applies various trained machine learning techniques on the captured image of the face and obtains a depth map and to apply one or more visual effects to the captured image. In some implementations, the AR/VR application continuously captures images of the user's face in real time or periodically to continuously or periodically update the applied one or more visual effects (e.g., the augmented reality eyewear or hat). This allows the user to move around in the real world and see the one or more visual effects update in real time.

(52) Data Architecture

(53) FIG. 3 is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the messaging server system **108**, according to certain examples. While the content of the database **126** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

(54) The database **126** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a

message, and included within the message data stored in the message table **302**, is described below with reference to FIG. **4**.

(55) An entity table **306** stores entity data, and is linked (e.g., referentially) to an entity graph **308** and profile data **316**. Entities for which records are maintained within the entity table **306** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

(56) The entity graph **308** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

(57) The profile data **316** stores multiple types of profile data about a particular entity. The profile data **316** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **316** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100**, and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

(58) Where the entity is a group, the profile data **316** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

(59) The database **126** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **304**) and images (for which data is stored in an image table **312**).

(60) The database **126** can also store data pertaining to individual and shared AR sessions. This data can include data communicated between an AR session client controller of a first client device **102** and another AR session client controller of a second client device **102**, and data communicated between the AR session client controller and the augmentation system **208**. Data can include data used to establish the common coordinate frame of the shared AR scene, the transformation between the devices, the session identifier, images depicting a body, skeletal joint positions, wrist joint positions, feet, and so forth.

(61) Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

(62) Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102**, or the current time.

(63) Other augmentation data that may be stored within the image table **312** includes augmented reality content items (e.g., corresponding to applying augmented reality experiences). An augmented reality content item or augmented reality item may be a real-time special effect and

sound that may be added to an image or a video.

(64) As described above, augmentation data includes augmented reality content items, overlays, image transformations, AR images, and similar terms that refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device **102** and then displayed on a screen of the client device **102** with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device **102** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. For example, multiple augmented reality content items that apply different pseudorandom movement models can be applied to the same content by selecting different augmented reality content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device **102** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

(65) Data and various systems using augmented reality content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two dimensional or three dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

(66) Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

(67) In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process

of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A first set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

(68) In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

(69) In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

(70) Other methods and algorithms suitable for face detection can be used. For example, in some examples, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

(71) In some examples, a search for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector is started. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

(72) A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

(73) In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a messaging client **104** operating on the client device **102**. The transformation system operating within the messaging client **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transformation system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the client device **102** as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

(74) The graphical user interface, presenting the modification performed by the transformation system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

(75) A story table **314** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **306**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

(76) A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client **104**, based on his or her location. The end result is a “live story” told from a community perspective.

(77) A further type of content collection is known as a “location story,” which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific

organization or other entity (e.g., is a student on the university campus).

(78) As mentioned above, the video table **304** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. Similarly, the image table **312** stores image data associated with messages for which message data is stored in the entity table **306**. The entity table **306** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **312** and the video table **304**.

(79) Data Communications Architecture

(80) FIG. 4 is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by a messaging client **104** for communication to a further messaging client **104** or the messaging server **118**. The content of a particular message **400** is used to populate the message table **302** stored within the database **126**, accessible by the messaging server **118**. Similarly, the content of a message **400** is stored in memory as “in-transit” or “in-flight” data of the client device **102** or the application servers **114**. A message **400** is shown to include the following example components: message identifier **402**: a unique identifier that identifies the message **400**. message text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the message **400**. message image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **312**. message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the video table **304**. message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the message **400**. message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data for a sent or received message **400** may be stored in the augmentation table **310**. message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the messaging client **104**. message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **406**, or a specific video in the message video payload **408**). message story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **314**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple images within the message image payload **406** may each be associated with multiple content collections using identifier values. message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition. message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** on which the message **400** was generated and from which the message **400** was sent. message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the message **400** is addressed.

(81) The contents (e.g., values) of the various components of message **400** may be pointers to

locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **312**. Similarly, values within the message video payload **408** may point to data stored within a video table **304**, values stored within the message augmentation data **412** may point to data stored in an augmentation table **310**, values stored within the message story identifier **418** may point to data stored in a story table **314**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **306**.

(82) True Size Estimation System

(83) FIG. 5 is a block diagram showing an example true size estimation system **224**, according to example examples. True size estimation system **224** includes a set of components **510** that operate on a set of input data (e.g., a monocular image depicting a face of a user **501**, depth map data **502**, and eyewear data **503**). True size estimation system **224** includes a landmark generation module **512**, a landmark selection module **514**, a depth map module **517**, a real-world scale computation module **516**, an image modification module **518**, an eyewear generation module **519**, and an image display module **520**. All or some of the components of the true size estimation system **224** can be implemented by a server, in which case, the monocular image depicting a face of the user **501** and the depth map data **502** are provided to the server by the client device **102**. In some cases, some or all of the components of the true size estimation system **224** can be implemented by the client device **102**.

(84) The landmark generation module **512** receives a monocular image depicting a face of a user **501**. This image can be received as part of a real-time video stream, a previously captured video stream or a new image captured by a camera of the client device **102**. The landmark generation module **512** applies one or more machine learning techniques to identify and segment a face of the user from the background of the monocular image depicting the face of the user **501**. The landmark generation module **512** can then apply one or more machine learning techniques to identify one or more landmarks on the identified face. For example, as shown in FIG. 6, the landmark generation module **512** can generate an output image **610** in which a plurality of landmarks **616**, **612** and **614** are provided. Each landmark uniquely and specifically identifies a region of the face, such as the eyebrows, the eyes, the nose, temple, the nose bridge, the mouth, the nose, the ears, the hair, the cheeks, the forehead, and so forth.

(85) The landmark generation module **512** can compute visibility and stability parameters for each of the identified landmarks. The visibility parameter specifies a score, rank or amount of the given landmark that is visible in the monocular image depicting a face of a user **501**. In an example, the landmark generation module **512** computes the visibility parameter by retrieving a generic three-dimensional (3D) facial model representation. The landmark generation module **512** identifies a set of landmarks on the 3D facial model representation. The landmark generation module **512** selects a given landmark from the identified landmarks of the user's face and matches the given landmark to a corresponding landmark on the 3D facial model. For example, the landmark generation module **512** selects the ears landmark and matches the ears landmark to the ears portion of the 3D facial model. The landmark generation module **512** computes how much of the ears landmark matches the ears portion of the 3D facial model to determine a level of visibility or visibility parameter. In an implementation, if 60 percent of the ears landmark matches the ears portion, the visibility parameter is set to a score of 60 percent. Namely, the visibility parameter is proportional to the amount of overlap or amount by which the landmark matches the corresponding portion of the 3D facial model.

(86) The landmark generation module **512** continues to compute the visibility parameter for the rest of the landmarks that are identified in the received image in a similar way using the 3D facial model. After obtaining the visibility parameter for each of the landmarks identified in the face, the landmark generation module **512** determines the visibility parameter of each landmark as a function of a number of the remaining set of landmarks that match the generic 3D facial model

representation. Specifically, the landmark generation module **512** computes how many of the landmarks have a visibility score that is greater than a certain threshold value (e.g., greater than 75 percent). The landmark generation module **512** then computes an overall visibility parameter for the identified landmarks based on the number of landmarks that have the visibility score that is greater than the threshold value. The greater the number of landmarks with the visibility score greater than the threshold value, the greater the overall visibility parameter.

(87) The landmark generation module **512** can also compute a stability parameter for each of the identified landmarks. The stability parameter indicates how much each given landmark moves over a threshold number of frames. For example, the landmark generation module **512** can receive a sequence of frames of a video that include the monocular image depicting a face of a user **501**. After identifying the landmarks on the face, the landmark generation module **512** can determine the 3D or 2D positions of the landmarks in the image. The landmark generation module **512** detects changes to the 3D or 2D positions of each landmark. Based on the amount of movement or changes to the 3D or 2D positions, the landmark generation module **512** generates and computes a corresponding stability parameter for the given landmark. Landmarks that have less amount of movement are assigned a greater stability parameter score than those that are determined to move more than a threshold amount per threshold number of frames (e.g., over 40 frames).

(88) In some cases, the landmark generation module **512** computes the stability parameter of each of the landmarks based on a frame rate associated with the client device **102** used to capture the video that depicts the face of the user. Namely, a client device **102** that has a first frame rate can be associated with a greater number of frames over which the stability parameter is computed than another client device with a lower second frame rate. Namely, the threshold number of frames can vary based on the frame rate of the client device **102**. After computing the stability and visibility parameters of each of the landmarks, the landmark generation module **512** sorts and ranks the landmarks based on their respective stability and visibility parameters.

(89) In some cases, prior to performing the visibility and stability parameter computations, the landmark generation module **512** can remove a set of interfering landmarks from the identified set of landmarks of the face. Namely, the landmark generation module **512** can access a predetermined list of known or predetermined set of interfering landmarks (e.g., a hair region, a facial garment (e.g., a face mask or eyeglasses), a neck region). In response to determining that one of the identified landmarks corresponds or is included among the predetermined set of interfering landmarks, the landmark generation module **512** can remove or discard or not consider such an identified landmark. After removing the set of interfering landmarks from the plurality of landmarks, the landmark generation module **512** provides a resulting set of remaining landmarks of the total identified plurality of landmarks to the landmark selection module **514**. In some cases, the removal of the interfering landmarks can be performed after computing the visibility and stability parameters of all of the identified plurality of landmarks on the face and before the top landmarks are selected. This way, even though a given landmark (e.g., a facial garment) is associated with a highest visibility and stability parameter than all other landmarks (e.g., the eyes and nose), the given landmark is not included or selected among the top landmarks by the landmark selection module **514** because it is considered to be an interfering landmark.

(90) The landmark generation module **512** provides the identified landmarks and their respective visibility and stability parameters to the landmark selection module **514**. The landmark selection module **514** selects a set of top landmarks that are associated with a greater visibility and greater stability parameters than a remaining set of landmarks. In some cases, the landmark selection module **514** can receive an indication of an augmented reality element that is to be included in the image that depicts the user's face. Based on the type of augmented reality element (e.g., eyewear augmented reality element), the landmark generation module **512** obtains a threshold quantity or number of landmarks that are to be selected as the top landmarks. For example, if the eyewear augmented reality element is selected, the threshold number of top landmarks includes two top

landmarks that need to be selected. In this case, the landmark selection module **514** obtains and tracks the two landmarks that are ranked higher than all other landmarks that are identified based on their respective visibility and stability parameters (or metrics). The threshold number of top landmarks can also be specified by the user.

(91) In some implementations, the landmark selection module **514** selects the top landmarks at random. For example, for a first sequence of frames or a first duration of a video, the landmark selection module **514** can select a first set of landmarks (e.g., eyes and nose) that have visibility and stability parameters that exceed a specified value. Then, for a second sequence of frames or a second duration of the video, the landmark selection module **514** can select a second set of landmarks (e.g., eyes and ears) that have visibility and stability parameters that exceed the specified value. Namely, the landmark selection module **514** can identify a collection of landmarks (e.g., more than the threshold number of top landmarks) which have visibility and stability parameters that are greater than a specified value or that satisfy a criterion or criteria. The landmark selection module **514** can then alternate and vary randomly which subset of the collection of landmarks (e.g., which two of the multiple collection of landmarks) to include in the threshold number of top landmarks.

(92) In an implementation, based on tracking the two landmarks, the landmark selection module **514** can provide the information pertaining to the two landmarks to the real-world scale computation module **516** to compute the real-world scale of the user depicted in the image. In this way, the real-world scale computation module **516** can compute a real-world scale of a user in a video or image based on a first set of landmarks at a first point in time (e.g., during a first portion of a video) and can compute the real-world scale of the user in the video or image based on a different second set of landmarks at a second point in time. The operations for identifying the facial landmarks, removing certain landmarks, and computing the stability and visibility parameters of the facial landmarks are repeated for each video frame or each subset of video frames to update the stability and visibility parameters. The updated stability and visibility parameters are provided to the landmark selection module **514** to update the landmarks selected as the top landmarks and iteratively correct the real-world scale of the face based on the updated plurality of landmarks.

(93) The depth map module **517** receives depth map data **502** from a depth sensor or depth camera of the client device **102**. The depth map data **502** is associated with the image or video being processed by the landmark generation module **512** and the landmark selection module **514**. The depth map module **517** receives a face segmentation mask for the face depicted in the monocular image depicting the face of the user **501**, such as from the landmark generation module **512**. The depth map module **517** applies momentum smoothing to the depth map data **502** based on the face segmentation mask. The depth map module **517** provides the smoothed depth map to the real-world scale computation module **516**. The real-world scale computation module **516** computes the real-world scale of the user's face depicted in the image based on the top landmarks provided by the landmark selection module **514** and the smoothed depth map received from the depth map module **517**. For example, the real-world scale computation module **516** can determine a distance to a given landmark from the client device **102** based on the smoothed depth map. Based on the distance, the real-world scale computation module **516** can compute the real-world facial measurements of the landmarks identified in the image received from the client device **102**. Namely, the real-world scale computation module **516** can use heuristics or machine learning techniques to compute the real-world physical size and measurements of the user's face when a landmark appears at a certain size and is at a certain distance away from the client device **102**. Using this information, the real-world scale computation module **516** can determine a scale that transforms the real-world physical measurements to the sizes of the landmarks and face depicted in the image captured by the client device **102**.

(94) The real-world scale computation module **516** provides the scale to the image modification module **518** and the eyewear generation module **519**. The eyewear generation module **519** obtains

an augmented reality graphical element that includes augmented reality eyewear. In an example, the real-world scale computation module **516** receives eyewear data **503**. The eyewear data **503** defines the physical material and rigid properties of physical sunglasses or eyeglasses. The eyewear data **503** also includes the physical measurements of the physical sunglasses or eyeglasses. The eyewear data **503** also provides an augmented reality element having a size and dimensions and behavior (deformation properties) that represents the physical sunglasses or eyeglasses.

(95) The eyewear data **503** can include information from a physical glasses designer that specifies physical attributes of the glasses, such as physical measurements, a style of the physical glasses, a lens shape and color, and a frame style and color.

(96) The eyewear generation module **519** can increase or decrease a size of the augmented reality element based on the real-world scale of the user's face provided by the real-world scale computation module **516**. For example, if the user's face in the image is at a first distance to the client device **102**, the scale can be determined based on the landmarks and smoothed depth data to be a first value. The eyewear generation module **519**, in this case, adjusts the size of the augmented reality element based on the first value. If the user's face is moved further away from the client device **102**, the face becomes smaller and is at a farther second distance. In this case, the scale can be determined to be a second smaller value than the first. In this case, the eyewear generation module **519** reduces the size of the augmented reality element based on the second value to be smaller.

(97) The eyewear generation module **519** provides the augmented reality element having the adjusted scale to the image modification module **518**. The image modification module **518** positions the scaled augmented reality element on the face of the user depicted in the image or video. In some cases, the image modification module **518** identifies a nose bridge landmark based on the output of the landmark generation module **512**. The image modification module **518** then identifies a nose bridge portion of the augmented reality element and centers the nose bridge portion of the augmented reality element on top of the nose bridge landmark. In some implementations, the image modification module **518** identifies a start point and end point of the nose bridge landmark. The image modification module **518** selects a point between the start and end points of the nose bridge landmark over which to position the nose bridge portion of the augmented reality element. The point between the start and end points can be heuristically determined, set by a user, or learned by a machine learning technique. In some cases, the image modification module **518** initially places the augmented reality element at a midpoint between the start and end points. For example, the image modification module **518** can determine a topology of the face of the user based on the landmarks generated by the landmark generation module **512**. The image modification module **518** can then position the scaled augmented reality graphical element within the image or video based on the topology of the user's face. In an example, the image modification module **518** positions a nose bridge portion of the augmented reality graphical element a predetermined distance above a nose bridge landmark within the topology.

(98) For example, as shown in FIG. 7, the image modification module **518** places the augmented reality eyewear **720** on the image **710** depicting the user's face. The nose bridge portion **722** of the augmented reality eyewear **720** is placed and positioned over the nose bridge landmark of the user's face.

(99) The image modification module **518** can detect a user's finger in the video stream received from the client device. The image modification module **518** can determine that the user's finger overlaps the nose bridge portion of the augmented reality element. The image modification module **518** can determine then that the finger is moved vertically towards the eyebrows. In response, the image modification module **518** adjusts the point at which the augmented reality element is positioned on the nose bridge portion by a specified amount to be closer to the eyebrows depending on the amount of movement of the finger. The image modification module **518** can determine then that the finger is moved vertically towards the mouth. In response, the image modification module

518 adjusts the point at which the augmented reality element is positioned on the nose bridge portion by a specified amount to be further from the eyebrows and closer to the mouth depending on the amount of movement of the finger. The image modification module **518** can record or store the position along the nose bridge landmark over which the nose bridge portion is placed. The image modification module **518** can place subsequent or other eyewear augmented reality elements such that their respective nose bridge portions are positioned at the stored position on the nose bridge landmark.

(100) The image modification module **518** can adjust the image captured by the camera and based on the output of the eyewear generation module **519**. The image modification module **518** adjusts the way in which the augmented reality eyeglasses or hat is positioned on a user(s) depicted in an image. Image display module **520** combines the adjustments made by the image modification module **518** into the received monocular image depicting the user's face. The image is provided by the image display module **520** to the client device **102** and can then be sent to another user or stored for later access and display.

(101) In an example, the eyewear generation module **519** can recommend or automatically select an augmented reality eyewear element to display on the user's face based on fit factors computed for a plurality of augmented reality eyewear elements. For example, the eyewear generation module **519** can receive, as part of the eyewear data **503**, a plurality of physical measurements of a plurality of physical glasses. The physical glasses can be associated with a particular manufacturer or can be a specified set of popular glasses or can include all of the physical glasses that have associated eyewear data **503**. The eyewear generation module **519** can select a subset of physical glasses for which to compute the fit factor based on a style and one or more attributes associated with the user, such as an age of the user, a gender of the user, preferences of the user, and the computed real-world scale of the face. Namely, the eyewear generation module **519** can determine that the physical size of the user's face is of a specified value and can select a subset of physical glasses that match the specified value. In another example, the eyewear generation module **519** can determine that the user in the image is a child and can select a set of child friendly glasses for which to compute the fit factor.

(102) The eyewear generation module **519** computes a fit factor for each of the physical glasses based on the physical measurements of the glasses provided by the eyewear data **503** and the real-world scale of the face of the user. In an implementation, the eyewear generation module **519** computes a fit factor for each of the glasses by determining a first distance between a nose bridge portion of each one of the plurality of physical glasses and a temples portion of the respective one of the plurality of physical glasses.

(103) The eyewear generation module **519** computes a second distance in the computed real-world scale of the face of the user between a nose bridge and a cheek bone or temple of the face of the user. Namely, the eyewear generation module **519** determines the real-world physical measurements of the landmarks depicted in the image of the user's face and the smoothed depth map and computes the distance between the nose bridge landmark and a cheek bone or temple landmark of the physical measurements of the face of the user. The eyewear generation module **519** then computes the fit factor for each of the glasses as a function of the first distance and the second distance. In an example, the fit factor represents a difference between the first and second distances, such that a smaller fit factor represents a better fit for the user. In some cases, the value for the fit factor is inversely related to the difference between the first and second distances, such that a greater fit factor represents a better fit for the user. Namely, the eyewear generation module **519** computes the fit factor for a given one of the glasses based on how close the distance between nose bridge portion of the given glasses and a temples portion of the given glasses is to the distance between the nose bridge landmark and a cheek bone or temple landmark of the user's face. The eyewear generation module **519** ranks and sorts all of the physical glasses based on the computed fit factor and automatically selects one or a subset of the physical glasses having a top fit factor

(e.g., a fit factor better than the rest) to present to the user.

(104) As another example, the eyewear generation module **519** computes a fit factor for each of the glasses by determining lens dimensions of each one of the plurality of physical glasses. For example, as shown in FIG. **8**, the eyewear generation module **519** determines the lens dimension **830** of the augmented reality glasses **820**. The lens dimension **830** shown in the image **810** of FIG. **8** is generated by scaling the lens dimension of the physical glasses based on the real-world scaling factor of the user's face. Namely, the lens dimension **830** is adjusted in size by an amount determined based on the real-world scaling factor of the user's face before being placed on the image **810** depicting the user's face. The eyewear generation module **519** determines the real-world physical measurements of the landmarks depicted in the image of the user's face and the smoothed depth map and then computes the fit factor for each of the glasses as a function of the lens dimensions and the real-world physical measurements of the landmarks. The eyewear generation module **519** ranks and sorts all of the physical glasses based on the computed fit factor and automatically selects one or a subset of the physical glasses having a top fit factor (e.g., a fit factor better than the rest) to present to the user.

(105) In some examples, the eyewear generation module **519** displays a warning message in response to determining that a fit factor for each of a plurality of physical glasses fails to be satisfied. For example, if the facial dimensions of the user exceed a measurement (e.g., are too small or too large) of each of the glasses (e.g., the distance between the nose bridge and temple on the user's face is larger by a specified amount than a distance between a nose bridge portion and temple of the glasses), the eyewear generation module **519** displays the warning message.

(106) In some examples, the eyewear generation module **519** displays a prompt shown in FIG. **9** that lists the top ranked glasses. Namely, the eyewear generation module **519** selects a specified number of glasses (e.g., three glasses) that have an associated fit factor that is better than fit factors associated with a remaining set of glasses. The eyewear generation module **519** can receive a user selection of physical glasses among the listed set of physical glasses. In response, the eyewear generation module **519** scales the augmented reality element representing the physical glasses selected by the user and positions the scaled augmented reality element within the image or video feed depicting the user's face.

(107) In some implementations, the image modification module **518** can deform the augmented reality element placed on top of the user's face in the image or video based on movement of the user's face. For example, the image modification module **518** can obtain rigid material properties associated with the augmented reality graphical element, such as by accessing the eyewear data **503**. The image modification module **518** deforms a first portion of the augmented reality graphical element in response to determining that the rigid material properties correspond to a first rigidity amount. The image modification module **518** deforms a plurality of portions of the augmented reality graphical element in response to determining that the rigid material properties correspond to a second rigidity amount. In some cases, the first rigidity amount is smaller than the second rigidity amount. For example, if the glasses are more rigid, the image modification module **518** can bend or deform the nose bridge portion of the augmented reality element and the temples portion of the augmented reality element when positioned on the user's face relative to the default formation of the augmented reality element when not placed on the user's face. As another example, if the glasses are more flexible, the image modification module **518** can bend or deform only the temples portion of the augmented reality element when positioned on the user's face relative to the default formation of the augmented reality element when not placed on the user's face.

(108) FIG. **10A** is a flowchart of a process **1000**, in accordance with some example examples. Although the flowchart can describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part,

may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.

(109) At operation **1001**, a client device **102** receives an image that includes a depiction of a face of a user, as discussed above. For example, the true size estimation system **224** can capture an image that depicts one or more faces of one or more users (e.g., a plurality of users).

(110) At operation **1002**, the client device **102** generates a plurality of landmarks of the face based on the received image, as discussed above. As an example, the true size estimation system **224** can generate the landmarks by applying a machine learning technique to the image.

(111) At operation **1003**, the client device **102** removes a set of interfering landmarks from the plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks, as discussed above.

(112) At operation **1004**, the client device **102** obtains a depth map for the face of the user, as discussed above.

(113) At operation **1005**, the client device **102** computes a real-world scale of the face of the user based on the depth map and the remaining set of landmarks, as discussed above.

(114) FIG. **10B** is a flowchart of a process **1010**, in accordance with some example examples.

Although the flowchart can describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part, may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.

(115) At operation **1011**, a client device **102** receives an image that includes a depiction of a face of a user, as discussed above.

(116) At operation **1012**, the client device **102** computes a real-world scale of the face of the user based on a selected subset of landmarks of the face of the user, as discussed above.

(117) At operation **1013**, the client device **102** obtains an augmented reality graphical element comprising augmented reality eyewear, as discussed above.

(118) At operation **1014**, the client device **102** scales the augmented reality graphical element based on the computed real-world scale of the face, as discussed above.

(119) At operation **1015**, the client device **102** positions the scaled augmented reality graphical element within the image on the face of the user.

(120) Machine Architecture

(121) FIG. **11** is a diagrammatic representation of the machine **1100** within which instructions **1108** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1100** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1108** may cause the machine **1100** to execute any one or more of the methods described herein. The instructions **1108** transform the general, non-programmed machine **1100** into a particular machine **1100** programmed to carry out the described and illustrated functions in the manner described. The machine **1100** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1100** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a

network bridge, or any machine capable of executing the instructions **1108**, sequentially or otherwise, that specify actions to be taken by the machine **1100**. Further, while only a single machine **1100** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1108** to perform any one or more of the methodologies discussed herein. The machine **1100**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1100** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

(122) The machine **1100** may include processors **1102**, memory **1104**, and input/output (I/O) components **1138**, which may be configured to communicate with each other via a bus **1140**. In an example, the processors **1102** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1106** and a processor **1110** that execute the instructions **1108**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **11** shows multiple processors **1102**, the machine **1100** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

(123) The memory **1104** includes a main memory **1112**, a static memory **1114**, and a storage unit **1116**, all accessible to the processors **1102** via the bus **1140**. The main memory **1104**, the static memory **1114**, and the storage unit **1116** store the instructions **1108** embodying any one or more of the methodologies or functions described herein. The instructions **1108** may also reside, completely or partially, within the main memory **1112**, within the static memory **1114**, within machine-readable medium **1118** within the storage unit **1116**, within at least one of the processors **1102** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1100**.

(124) The I/O components **1138** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1138** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1138** may include many other components that are not shown in FIG. **11**. In various examples, the I/O components **1138** may include user output components **1124** and user input components **1126**. The user output components **1124** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1126** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(125) In further examples, the I/O components **1138** may include biometric components **1128**,

motion components **1130**, environmental components **1132**, or position components **1134**, among a wide array of other components. For example, the biometric components **1128** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1130** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

(126) The environmental components **1132** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

(127) With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

(128) Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

(129) The position components **1134** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

(130) Communication may be implemented using a wide variety of technologies. The I/O components **1138** further include communication components **1136** operable to couple the machine **1100** to a network **1120** or devices **1122** via respective coupling or connections. For example, the communication components **1136** may include a network interface component or another suitable device to interface with the network **1120**. In further examples, the communication components **1136** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), WiFi® components, and other communication components to provide communication via other modalities. The devices **1122** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

(131) Moreover, the communication components **1136** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1136** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC

RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1136**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

(132) The various memories (e.g., main memory **1112**, static memory **1114**, and memory of the processors **1102**) and storage unit **1116** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1108**), when executed by processors **1102**, cause various operations to implement the disclosed examples.

(133) The instructions **1108** may be transmitted or received over the network **1120**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1136**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1108** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1122**.

(134) Software Architecture

(135) FIG. **12** is a block diagram **1200** illustrating a software architecture **1204**, which can be installed on any one or more of the devices described herein. The software architecture **1204** is supported by hardware such as a machine **1202** that includes processors **1220**, memory **1226**, and I/O components **1238**. In this example, the software architecture **1204** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1204** includes layers such as an operating system **1212**, libraries **1210**, frameworks **1208**, and applications **1206**. Operationally, the applications **1206** invoke API calls **1250** through the software stack and receive messages **1252** in response to the API calls **1250**.

(136) The operating system **1212** manages hardware resources and provides common services. The operating system **1212** includes, for example, a kernel **1214**, services **1216**, and drivers **1222**. The kernel **1214** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1214** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **1216** can provide other common services for the other software layers. The drivers **1222** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1222** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

(137) The libraries **1210** provide a common low-level infrastructure used by the applications **1206**. The libraries **1210** can include system libraries **1218** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1210** can include API libraries **1224** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1210** can also include a wide variety of other libraries **1228** to provide many other APIs to the applications **1206**.

(138) The frameworks **1208** provide a common high-level infrastructure that is used by the applications **1206**. For example, the frameworks **1208** provide various graphical user interface

(GUI) functions, high-level resource management, and high-level location services. The frameworks **1208** can provide a broad spectrum of other APIs that can be used by the applications **1206**, some of which may be specific to a particular operating system or platform.

(139) In an example, the applications **1206** may include a home application **1236**, a contacts application **1230**, a browser application **1232**, a book reader application **1234**, a location application **1242**, a media application **1244**, a messaging application **1246**, a game application **1248**, and a broad assortment of other applications such as an external application **1240**. The applications **1206** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1206**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the external application **1240** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the external application **1240** can invoke the API calls **1250** provided by the operating system **1212** to facilitate functionality described herein.

Glossary

(140) “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

(141) “Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

(142) “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

(143) “Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component

may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function or related functions.

(144) Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

(145) A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

(146) Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.

(147) Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(148) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or

permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1102** or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

(149) “Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

(150) “Ephemeral message” refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

(151) “Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

(152) “Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

(153) “Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

(154) Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

Claims

1. A method comprising: removing, from an object depicted in an image, a set of interfering landmarks from a plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks; obtaining a three-dimensional object representation; matching each of the remaining set of landmarks to the three-dimensional object representation; determining, for each of the remaining set of landmarks, a visibility parameter based on the three-dimensional object representation, each visibility parameter indicating a degree of visual detectability of each landmark of the plurality of landmarks in the three-dimensional object representation; determining, for each of the remaining set of landmarks, a stability parameter as a function of movement of the remaining set of landmarks over a plurality of frames of a video comprising the image, the stability parameter indicating how much each of the remaining set of landmarks moves over a threshold number of frames of the plurality of frames and being computed by detecting changes to positions of each of the remaining set of landmarks over the threshold number of frames, landmarks having less amount of movement being assigned a greater stability parameter score than landmarks that are determined to move more than a threshold amount over the threshold number of frames; sorting the remaining set of landmarks based on both the visibility parameter of each of the remaining set of landmarks and the stability parameter of each of the remaining set of landmarks; and computing a real-world scale of the object depicted in the image based on landmarks selected from the sorted remaining set of landmarks.
2. The method of claim 1, the object comprising a face of a user, further comprising: generating the plurality of landmarks of the face based on the image; and obtaining a depth map for the face of the user, the real-world scale of the face being computed based on the depth map.
3. The method of claim 1, wherein the threshold number of frames is determined based on a frame rate of the video, higher frame rates corresponding to a greater threshold number of frames than lower frame rates.
4. The method of claim 3, further comprising: tracking movement of the remaining set of landmarks over the plurality of frames; computing the stability parameter based on a frame rate of the video; and varying the threshold number of frames based on the frame rate of the video.
5. The method of claim 1, wherein determining the visibility parameter comprises: computing how much of each of the remaining set of landmarks matches a corresponding portion of the three-dimensional object representation; and setting the visibility parameter for each of the remaining set of landmarks proportional to an amount by which each landmark matches the corresponding portion of the three-dimensional object representation.
6. The method of claim 1, wherein the set of interfering landmarks comprises at least one of a hair region, one or more facial garments, or a face mask.
7. The method of claim 1, further comprising applying a machine learning model to the image to identify the plurality of landmarks.
8. The method of claim 1, further comprising: selecting a threshold number of top landmarks from the remaining set of landmarks based on visibility or stability parameters associated with each of the remaining set of landmarks, wherein the real-world scale of the object is computed based on the selected threshold of top landmarks.
9. The method of claim 8, wherein the threshold number comprises two top landmarks.
10. The method of claim 8, further comprising: selecting eyes and nose landmarks as the top landmarks in response to determining that the eyes and nose landmarks are associated with greater visibility and stability parameters than other landmarks in the remaining set of landmarks.

11. The method of claim 8, wherein the top landmarks are selected at random from the remaining set of landmarks by: identifying a collection of landmarks having visibility and stability parameters that exceed a specified value; and alternating randomly between different subsets of the collection of landmarks during different portions of the video.
12. The method of claim 11, wherein a first set of top landmarks are selected for a first subset of the frames, and wherein a second set of top landmarks are selected for a second subset of frames.
13. The method of claim 1, further comprising: generating an object segmentation mask for the object depicted in the image; and applying momentum smoothing to a depth map based on the object segmentation mask.
14. The method of claim 1, further comprising: updating the plurality of landmarks as each frame of the video depicting the object is received; and iteratively correcting the real-world scale of the object based on the updated plurality of landmarks by repeating the removing and obtaining operations for the updated plurality of landmarks.
15. The method of claim 1, further comprising: obtaining an augmented reality graphical element comprising augmented reality eyewear; computing fit factors for a plurality of augmented reality eyewear elements based on physical measurements of corresponding glasses; ranking the plurality of augmented reality eyewear elements based on the computed fit factors; selecting the augmented reality graphical element based on the ranking; identifying a nose bridge landmark based on the remaining set of landmarks; and positioning the augmented reality graphical element within the image on the object based on the nose bridge landmark.
16. The method of claim 15, further comprising: detecting a user's finger overlapping the nose bridge landmark; adjusting a vertical position of the augmented reality graphical element based on movement of the user's finger; and positioning the nose bridge landmark of the augmented reality graphical element above the nose bridge landmark based on the vertical position.
17. The method of claim 15, further comprising: obtaining rigid material properties associated with the augmented reality graphical element; deforming the augmented reality graphical element based on the rigid material properties; and adjusting a scale of the augmented reality graphical element based on the computed real-world scale of the object.
18. The method of claim 1, further comprising: computing a distance between the remaining set of landmarks; retrieving a measure of depth for the remaining set of landmarks; and generating a scaling factor based on the distance and the measured depth that relates a size of the object in the image to a real-world size of the object, wherein a size of an augmented reality graphical element is modified as a function of the scaling factor.
19. A system comprising: at least one processor; and a memory component having instructions stored thereon, when executed by the processor, causes the processor to perform operations comprising: removing, from an object depicted in an image, a set of interfering landmarks from a plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks; obtaining a three-dimensional object representation; matching each of the remaining set of landmarks to the three-dimensional object representation; determining, for each of the remaining set of landmarks, a visibility parameter based on the three-dimensional object representation, each visibility parameter indicating a degree of visual detectability of each landmark of the plurality of landmarks in the three-dimensional object representation; determining, for each of the remaining set of landmarks, a stability parameter as a function of movement of the remaining set of landmarks over a plurality of frames of a video comprising the image, the stability parameter indicating how much each of the remaining set of landmarks moves over a threshold number of frames of the plurality of frames and being computed by detecting changes to positions of each of the remaining set of landmarks over the threshold number of frames, landmarks having less amount of movement being assigned a greater stability parameter score than landmarks that are determined to move more than a threshold amount over the threshold number of frames; sorting the remaining set of landmarks based on both the visibility parameter of each of the remaining set of landmarks and the

stability parameter of each of the remaining set of landmarks; and computing a real-world scale of the object depicted in the image based on landmarks selected from the sorted remaining set of landmarks.

20. A non-transitory computer-readable storage medium having stored thereon, instructions when executed by a processor, causes the processor to perform operations comprising: removing, from an object depicted in an image, a set of interfering landmarks from a plurality of landmarks resulting in a remaining set of landmarks of the plurality of landmarks; obtaining a three-dimensional object representation; matching each of the remaining set of landmarks to the three-dimensional object representation; determining, for each of the remaining set of landmarks, a visibility parameter based on the three-dimensional object representation, each visibility parameter indicating a degree of visual detectability of each landmark of the plurality of landmarks in the three-dimensional object representation; determining, for each of the remaining set of landmarks, a stability parameter as a function of movement of the remaining set of landmarks over a plurality of frames of a video comprising the image, the stability parameter indicating how much each of the remaining set of landmarks moves over a threshold number of frames of the plurality of frames and being computed by detecting changes to positions of each of the remaining set of landmarks over the threshold number of frames, landmarks having less amount of movement being assigned a greater stability parameter score than landmarks that are determined to move more than a threshold amount over the threshold number of frames; sorting the remaining set of landmarks based on both the visibility parameter of each of the remaining set of landmarks and the stability parameter of each of the remaining set of landmarks; and computing a real-world scale of the object depicted in the image based on landmarks selected from the sorted remaining set of landmarks.
